

Session-02: Discrete probability distribution

1. The probability density function of a random variable X is

X	0	1	2	3	4	5	6
$P(X)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

Find $P(X < 4)$; $P(X \geq 5)$; $P(3 < X \leq 6)$. What will be the minimum value of k so that $P(X \leq 2) > 0.3$.

Answer: $k = \frac{1}{49}$; $P(X < 4) = \frac{16}{49}$; $P(X \geq 5) = \frac{24}{49}$; $P(3 < X \leq 6) = \frac{33}{49}$;

For $P(X \leq 2)$, $k = \frac{1}{30}$

2. A coin is tossed three times. Let X denote the number of heads showing up. Find the distribution of X . Also, find its mean and variance.

Answer:

X	0	1	2	3
$P(X)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Mean = $\frac{3}{2}$; Variance = $\frac{3}{4}$